

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-ae2963eef6bb0f3b3.jsonl`  
- SHA-256: `0ac2c0d7144379ed2ce5f5bf7711cd4869cb530f2ff0fe960a2dc3dadad6c0ed`  
- Source modified: `2026-06-06T17:32:21+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:30:52.246Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

****URL:**** https://openaccess.thecvf.com/content/CVPR2024W/CVsports/papers/Xarles_T-DEED_Temporal-Discriminability_Enhancer_Encoder-Decoder_for_Precise_Event_Spotting_in_Sports_CVPRW_2024_paper.pdf
****Title:**** T-DEED: Temporal-Discriminability Enhancer Encoder-Decoder for Precise Event Spotting in Sports (CVPRW 2024)
****Found via:**** recent/applied search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:54.508Z)

I'll fetch the source and extract relevant claims.

3. user (2026-06-06T17:31:18.333Z)

T-DEED: Temporal-Discriminability Enhancer Encoder-Decoder - Key Claims

1. Task and Method

****Task:**** Precise event spotting in sports videos through temporal event localization.

****Method Architecture:**** The paper proposes an encoder-decoder framework that enhances temporal discriminability. The approach uses "Temporal-Discriminability Enhancer" mechanisms to improve event localization precision beyond standard frame-level classification.

2. Datasets

Based on the PDF metadata, datasets likely include sports video benchmarks commonly used for event spotting, though specific dataset names aren't clearly extracted from the available content sections.

3. Performance Metrics

Specific accuracy/mAP numbers and tolerance thresholds are embedded in the compressed PDF content and not readily extractable from the visible sections.

4. Compute and Architecture

The specific backbone architecture, GPU requirements, parameter counts, and inference costs are contained within the compressed document sections and cannot be reliably extracted from the provided text.

5. License Information

No explicit license information for code, weights, or datasets is visible in the extracted content.

6. Publication Details

****Venue:**** "IEEE Conference on Computer Vision and Pattern Recognition Workshops"

****Authors:**** Artur Xarles, Sergio Escalera, Thomas B. Moeslund, Albert Clapes

****Processing Date:**** The document was processed on 2024-06-02.

***Note:** The PDF content is heavily compressed, limiting detailed extraction of numerical results and technical specifications.*

[Binary content (application/pdf, 5.3MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767069364-9guyic.pdf]

4. assistant (2026-06-06T17:31:21.313Z)

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5. user (2026-06-06T17:31:22.206Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

6. assistant (2026-06-06T17:31:25.193Z)

Let me check for a PDF text-extraction tool available on the system.

7. user (2026-06-06T17:31:27.362Z)

Exit code 1

Traceback (most recent call last):

```
File "<string>", line 1, in <module>
    import pdfplumber; print('pdfplumber ok')
^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

ModuleNotFoundError: No module named 'pdfplumber'

Traceback (most recent call last):

```
File "<string>", line 1, in <module>
    import fitz; print('pymupdf ok')
^^^^^^^^^^^^^^^^
```

ModuleNotFoundError: No module named 'fitz'

Traceback (most recent call last):

```
File "<string>", line 1, in <module>
    import PyPDF2; print('pypdf2 ok')
^^^^^^^^^^^^^^^^
```

ModuleNotFoundError: No module named 'PyPDF2'

Traceback (most recent call last):

```
File "<string>", line 1, in <module>
    import pypdf; print('pypdf ok')
^^^^^^^^^^^^^^^^
```

ModuleNotFoundError: No module named 'pypdf'

8. user (2026-06-06T17:31:35.813Z)

[notice] A new release of pip is available: 26.0.1 -> 26.1.2

[notice] To update, run: python.exe -m pip install --upgrade pip

9. user (2026-06-06T17:31:43.702Z)

PAGES: 10

==== PAGE 1 ====

T-DEED: Temporal-Discriminability Enhancer Encoder-Decoder for Precise Event Spotting in Sports Videos

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Abstract

In this paper, we introduce T-DEED, a Temporal-

Discriminability Enhancer Encoder-Decoder for Precise Event Spotting in sports videos. T-DEED addresses multiple challenges in the task, including the need for discriminability among frame representations, high output temporal resolution to maintain prediction precision, and the necessity to capture information at different temporal scales to handle events with varying dynamics. It tackles these challenges through its specifically designed architecture, featuring an encoder-decoder for leveraging multiple temporal scales and achieving high output temporal resolution, along with temporal modules designed to increase token discriminability. Leveraging these characteristics, T-DEED achieves SOTA performance on the FigureSkating and FineDiving datasets. Code is available at <https://github.com/arturxe2/T-DEED>.

1. Introduction

In recent years, remarkable progress has been made in the field of video understanding, largely driven by advancements in deep learning and increased computational power. These developments have enabled researchers to move beyond simpler tasks like action recognition [18] in trimmed videos to more complex challenges such as accurate localization of actions within untrimmed videos. Among these tasks, we find Temporal Action Localization (TAL), which represents actions as temporal intervals, and Action Spotting (AS), which uses single keyframes to represent them. While TAL [37] has historically received more attention, AS has recently experienced an increase in research interest, particularly in domains such as sports, as fast-paced actions ? common in sports videos ? can be better represented by single temporal positions rather than intervals of time. Additionally, Hong et al. [20] extended AS to Precise Event Spotting (PES), distinguishing events from actions and employing tighter tolerances for evaluation.

This paper specifically focuses on the task of Precise Event Spotting in sports (illustrated in Figure 1), evaluated on FigureSkating [19] and FineDiving [38], two common sports datasets. Following Hong et al. [20], we use tight tolerances to accommodate the fast-paced nature of sporting events, where even a minor temporal deviation of 1-2 frames can lead to missed events. Three main challenges faced by methods addressing this task include: (1) the need for discriminative per-frame representations to differentiate between adjacent frames with high spatial similarity, (2) the necessity of high output temporal resolution to avoid losing prediction precision, and (3) the variability in the amount of temporal context required for different events, influenced by dataset characteristics and event dynamics.

To tackle these challenges we introduce T-DEED, a Temporal-Discriminability Enhancer Encoder-Decoder for PES in sports videos. Introducing skip connections within its encoder-decoder architecture, T-DEED operates across various temporal scales, capturing actions requiring diverse temporal contexts, while restoring the original temporal resolution, thereby addressing challenges (2) and (3). Additionally, it integrates Scalable-Granularity Perception (SGP) [31] based layers to increase discriminability among features within the same temporal sequence, tackling challenge (1). To summarize, our main contributions are:

1. We incorporate residual connections into the SGP layer, enabling the integration of features from multiple temporal scales within the skip connections of our encoder-decoder architecture. This results in the creation of the SGP-Mixer layer, addressing challenge (3).
2. We introduce the SGP-Mixer module within the SGP-

Mixer layer, which adapts the SGP module to aggregate information from different temporal scales. This module shares the core principles of SGP to enhance token discriminability while modeling temporal information, therefore tackling challenge (1).

This CVPR Workshop paper is the Open Access version, provided by the Computer Vision Foundation. Except for this watermark, it is identical to the accepted version; the final published version of the proceedings is available on IEEE Xplore.

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==== PAGE 2 ====

Figure 1. Illustration of the Precise Event Spotting task on the FigureSkating dataset. Red-marked frames contain events that require precise localization and correct classification among possible classes.

3. We conduct extensive ablations of T-DEED components, highlighting the advantages of the encoder-decoder architecture with SGP-based layers to enhance token discriminability. Additionally, T-DEED achieves state-of-the-art performance. Specifically, on FigureSkating, it improves mean Average Precision (mAP) by +1.15 for the FS-Comp split and +3.07 for the FS-Perf split. Similarly, on the FineDiving dataset, T-DEED achieves an improvement of +4.83 mAP.

2. Related work

Over the past decade, deep learning has driven the field of video understanding through a remarkable evolution. Initially focused on simple tasks like classifying short-trimmed videos [1, 6, 16], the field has transitioned to more complex challenges, including Temporal Action Localization (TAL) and Action Spotting (AS). Both TAL and AS focus on temporally locating specific actions within untrimmed videos. TAL specifies temporal intervals for annotations, while AS represents actions with single keyframes, making TAL suitable for prolonged actions and AS more appropriate for fast-occurring actions, with the added benefit of reduced annotation costs. Furthermore, Hong et al. [20] extended the AS task to Precise Event Spotting (PES), introducing a key difference in the required precision of predictions, limited to only a few frames, and distinguishing between actions and events. In the sports domain, AS and PES are particularly popular as they adapt better to its fast nature, leading to the development of many approaches [8, 11, 15, 20, 32, 36] across different sports, including football [10, 14], figure skating [19], diving [38], gymnastics [30], and tennis [45]. Given the inherent similarities between TAL and AS, the methodologies developed for these tasks frequently share common components. However, TAL methods have attracted more attention due to the earlier introduction of the task and a more extensive set of benchmarking datasets [5, 9, 22, 42, 46], placing them a step ahead of AS methods. In contrast, AS methods have been tailored for specific datasets and competitive challenges, exemplified by their development in challenges like SoccerNet Action Spotting and SoccerNet Ball Action Spotting [8, 15]. Notably, E2E-Spot [20] stands out as the only action or event spotting method evaluated across multiple datasets.

Temporal Action Localization. In TAL methods, a common classification divides them into two groups: two-stage methods [3, 13, 17, 28, 39] and one-stage methods [23, 25, 31, 41, 43]. Two-stage methods generate class-agnostic proposals that are later classified into action labels or background, while one-stage models directly localize and classify actions in a single step, offering simplicity and achieving state-of-the-art performance in many TAL and AS scenarios.

Among one-stage methods, early approaches [4, 24] utilized anchor windows to generate action predictions. Later, Yang et al. [40] introduced an anchor-free approach, relying on temporal points instead of anchor windows, highlighting the advantages of both techniques. Building on this approach, current state-of-the-art methods such as ActionFormer [43] and TriDet [31] have exhibited remarkable performance across various datasets. They leverage a feature pyramid network to process features at different temporal scales, a critical aspect for identifying actions that require distinct temporal contexts. The main difference lies in their prediction head and the layers used for feature processing. ActionFormer employs transformer layers, later revealed to suffer from the rank loss problem [12], negatively impacting token discriminability. To alleviate this problem, TriDet proposes a more efficient convolutional-based Scalable-Granularity Perception (SGP) layer, specially designed to increase token discriminability within the same temporal sequence, contributing to an improved overall performance. T-DEED, inspired by TAL trends, focuses on enhancing token discriminability while leveraging multiple temporal scales, with modifications tailored to meet the precision requirements of PES.

Action Spotting. In AS, techniques similar to those in TAL have demonstrated state-of-the-art performance on the SoccerNet challenges [8, 15]. Many methods [11, 32, 36] classify temporal points as either background or actions and refine them through temporal regression using either convolutional [32] or Transformer-based [11, 36] approaches. In contrast, Hong et al. [20] propose a simple end-to-end solution that employs a convolutional backbone with Gate-Shift Modules (GSM) [33] for extracting per-frame features with short-term temporal information, followed by a Gated Recurrent Unit (GRU) [7] layer for long-term temporal in-

==== PAGE 3 ====

Figure 2. Illustration of T-DEED architecture comprising three key components: (1) Feature extractor to produce per-frame representations, (2) Temporally discriminant encoder-decoder to capture local and global temporal information while enhancing token discriminability, and (3) Prediction head to generate per-frame classifications and displacements for refinement.

formation. This approach proves effective in their proposed task of PES across four different datasets [19, 30, 38, 45], and is also adaptable to the coarser AS task in SoccerNet. While many TAL and AS methods [31, 32, 36, 43] rely on pre-extracted features due to their efficiency in training, end-to-end approaches have demonstrated that they can be beneficial in learning more meaningful features in some cases. This is exemplified by Hong et al. [20], particularly in scenarios where precise predictions are essential, such as in the case of PES. Exploiting these advantages, T-DEED also adopts an end-to-end approach.

3. Method

Problem definition. Precise Event Spotting (PES) involves the identification and localization of actions within an untrimmed video X , as illustrated in Figure 1. Given the video input, the objective is to recognize and locate all the events occurring in the video, represented as $E = \{e_1, \dots, e_N\}$. The number of events, denoted as N , may vary across different videos. Each event instance e_i comprises an action class $c_i \in \{1, \dots, C\}$ (where C is the number of distinct event classes) and its corresponding temporal position t_i (i.e. the exact frame where it occurs), forming a

pair $e_i = (c_i, t_i)$.

Method overview. Our model, Temporal-Discriminability Enhancer Encoder-Decoder (T-DEED), is designed to increase token discriminability for Precise Event Spotting (PES) while leveraging multiple temporal scales. As illustrated in Figure 2, T-DEED comprises three main blocks: a feature extractor, a temporally discriminant encoder-decoder, and a prediction head. We process videos through fixed-length clips, each containing L densely sampled frames. The feature extractor, composed of a 2D backbone with Gate-Shift-Fuse (GSF) modules [34], handles the input frames, generating per-frame representations of dimension d , hereby referred to as tokens. These tokens undergo further refinement within the temporally discriminant encoder-decoder. This module employs SGP layers which, as shown by Shi et al. [31], diminish token similarity, thereby boosting discriminability across tokens of the same sequence. The encoder-decoder architecture allows the processing of features across diverse temporal scales, helpful for detecting events requiring different amounts of temporal context. We also integrate skip connections to preserve the fine-grained information from the initial layers. To effectively merge information proceeding from varying temporal scales in the skip connections, we introduce the SGP-Mixer layer, detailed in Section 3.2. This layer employs the same principles as the SGP layer to enhance token discriminability while gathering information from varying range temporal windows. Finally, the output tokens are directed to the prediction head, resembling those commonly used in Action Spotting (AS) literature [11, 32, 36]. It encompasses a classification component to identify whether an event occurs at the given temporal position or in close proximity (within a radius of r_E frames). Additionally, for the positive classifications, a displacement component pinpoints the exact temporal position of ground truth events. Further details of the proposed method are discussed in the following sections.

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==== PAGE 4 ====

3.1. Feature extractor

The feature extractor processes the input frame sequence, $RL \times H \times W \times 3$ with $H \times W$ denoting the spatial resolution, and produces per-frame feature representations, $RL \times d$. Following Hong et al. [20], we employ a compact 2D backbone to ensure efficiency and to accommodate longer video sequences. For comparability with previous PES methods, we choose RegNetY [29] as our feature extractor, known for its efficiency. To incorporate local temporal information during the extractor, we utilize GSF [34], which outperforms GSM [33] in action recognition tasks. GSF modules are specifically applied to the latter half of the RegNetY backbone, enabling local spatial modelling before integrating longer-term temporal context.

3.2. Temporally discriminant encoder-decoder

The temporally discriminant encoder-decoder processes the tokens produced by the feature extractor module, which contain mainly spatial information, with the objective of enriching them with essential temporal information, both local and global, to enable precise event predictions. To accomplish this, it incorporates three key components: (1) an encoder-decoder architecture for exploiting information across diverse temporal scales, (2) SGP layers to increase discriminability among tokens within the same sequence,

and (3) our novel SGP-Mixer layer, designed to effectively fuse features from varying temporal scales while maintaining the distinctiveness of its output tokens.

Encoder-decoder architecture. Various works [31, 32, 43] have shown the benefits of processing features at different temporal scales in tasks related to action recognition. This is because certain actions inherently require longer temporal context for recognition and localization, while others can be identified with just a few frames. SOTA models in TAL achieve this by employing a feature pyramid, where the temporal dimension is downsampled by a factor of k in the final layers through max-pooling, and predictions are made for each output feature at various temporal scales. However, in PES, where precision is crucial, this is detrimental. As shown in Section 4.5, the further down we go in the feature pyramid, the more diminishes the precision of the output predictions, inevitably impacting model performance. To address this issue while still leveraging different temporal scales, we propose an encoder-decoder architecture. After temporal downsampling during the encoder, akin to the feature pyramid, we restore the original temporal resolution through the decoder, thereby regaining frame-level granularity for the representations. We do this by incorporating skip connections from the encoder to the decoder’s upsampling.

Specifically, the encoder begins by complementing the tokens with a learnable encoding that specifies its temporal position. Furthermore, it comprises B encoder blocks, each consisting of an SGP layer to enhance discriminability while capturing temporal context, and a max-pooling operation to reduce the temporal dimension by a factor of k . An additional SGP layer is applied in the neck of the encoder-decoder, before passing the features to the decoder. In the decoder, the original temporal resolution is restored through B decoder blocks. Each of these blocks incorporates an SGP-Mixer layer, which extends the SGP layer to increase the temporal dimension by a factor of k while integrating information coming from the skip connections.

SGP layer. The SGP layer, as introduced by Shi et al. [31], addresses the rank-loss problem [12] commonly encountered in Transformers [35], thus improving token discriminability within sequences. As depicted in the top of Figure 3, it replaces the self-attention module in the Transformer layer with an SGP module illustrated at the bottom of the figure. This module comprises two primary branches: an instant-level branch and a window-level branch. The instant-level branch aims to boost token discriminability by increasing its distance from the clip-level average token, while the window-level branch captures temporal information from multiple receptive fields. Furthermore, the SGP layer substitutes one of the layer normalization modules with group normalization.

Given its advantageous characteristics, which contributed to Tridet’s SOTA results in TAL, we believe it can be even more beneficial in PES, where precise frame-level predictions are crucial and can benefit from improved discriminability between concurrent tokens, as discussed in Section 4.5.

SGP-Mixer layer. As shown in the top of Figure 4, the SGP-Mixer layer extends the original SGP layer to accommodate two inputs with distinct temporal scales. This adaptation becomes necessary in our architecture due to the reception of an input feature $\mathbf{z} \in \mathbb{R}^{L/k \times d}$ from the preceding layer, and features $\mathbf{x} \in \mathbb{R}^{L/(j \cdot 1) \times d}$ from the skip connec-

tion, where $j \in \{1, \dots, B\}$ denotes the depth within the decoder blocks. Both features undergo layer normalization before entering the SGP-Mixer module, where a combination of both features is generated. Finally, the output features are further processed through identical components within an SGP layer.

The fusion of both features is done in the SGP-Mixer module, shown at the bottom of the Figure 4. First, it up-samples the features from the previous layer to match the temporal dimensions (RL/k_j $\rightarrow RL/k(j+1)$) using linear interpolation. Following the principles of the SGP layer, it has two instant-level branches – one for each input feature – to improve token discriminability. This is particularly im-

10. assistant (2026-06-06T17:31:48.638Z)

I have substantial content. Let me extract the experimental results pages (datasets, mAP numbers, tolerances, compute) to get the falsifiable details.

11. user (2026-06-06T17:31:55.581Z)

==== PAGE 5 ====

Figure 3. Illustration of the SGP layer structure, with its module comprising an instant-level branch to boost token discriminability and a window-level branch for temporal modeling.

portant for the features in the previous layer, since upsampling may produce similar tokens in adjacent temporal positions. In addition, it includes two window-level branches to aggregate information from both features while capturing different temporal contexts. In each branch, one feature evolves to mix information across different temporal receptive fields, which is later gated by the other feature. Shortcuts are also introduced, but, unlike SGP, feature aggregation from different branches involves concatenation and linear projection, which we find more effective than simple addition, as will be shown later in Section 4.5.

3.3. Prediction head

Following common AS approaches [11, 32, 36], we include a prediction head consisting of a classification head and a displacement head. The classification head uses a linear layer and a softmax activation to project the output of the temporally discriminant encoder-decoder, $RL \times d$, onto $\mathbb{R}^C$, representing the probability of each temporal position containing each of the events or a background class. Similarly, the displacement head uses a linear layer to project the same output to $\mathbb{R}^1$, representing the displacement toward the ground truth event if an actual event is present at the corresponding temporal position.

3.4. Training details

The model is trained using a combination of a classification loss (L_c) and a displacement loss (L_d). For classification, per-frame cross-entropy loss is employed weighting the positive classes by a factor of w for all events. Mean squared error is used for displacement. The final loss (L) for a given clip is the sum of both losses:

Figure 4. Illustration of the SGP-Mixer layer structure integrating an SGP-Mixer module to aggregate features of different temporal scales. This module follows the SGP principles, incorporating instant-level and window-level branches to boost token discriminability while merging the features.

$$L = L_c + L_d$$

L

LX
 $l=1$
 $CE(w(y_c$
 l, y_c
 $l) + M SE(y_d$
 l, y_d
 $l), (1)$
where y_c
 l represents the one-hot encoding of the event in
frame l , y_c
 l denotes the classification probabilities at that
frame, y_d
 l indicates the actual displacement to a ground-
truth event (if an event is within the detection radius r_E),
and y_d
 l represents the predicted displacement.
During training, we perform typical data augmentation
techniques such as random cropping, random horizontal
flip, gaussian blur, color jitter, and mixup [44].

3.5. Inference

At inference time, the data augmentation techniques are dis-
abled and we use clips with 50% of overlapping. More-
over, to reduce the number of candidate events, Soft Non-
Maximum Suppression [2] is applied.

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Table 1. Comparison with SOTA methods on FigureSkating (FS-Comp and FS-Perf splits) and
FineDiving datasets, highlighting the best
results in bold and the second-top performance with underlining. For each model, we report
feature extractor sizes in MegaFlops (MF) and
input modalities including RGB images and Optical Flow (OF).

Model	Size	Input	? = 1	2	1	2	1	2
E2E-Spot [20]	200MF	RGB	81.00	93.50	85.10	95.70	68.40	85.30
E2E-Spot [20]	800MF	RGB	84.00	-	83.60	-	64.60	-
E2E-Spot [20]	800MF	RGB + OF [21]	83.40	94.90	83.30	96.00	66.40	84.80
T-DEED	200MF	RGB	85.15	91.70	86.79	96.05	71.48	87.62
T-DEED	800MF	RGB	84.77	92.86	88.17	95.87	73.23	88.88

4. Results

In this section, we detail the evaluation setup for our pro-
posed method and present experimental results demonstrat-
ing its superiority over current SOTA methods in PES on the
application domain of sports data. Additionally, we conduct
ablations on key components of our proposal.

4.1. Datasets

We conduct experiments on two challenging datasets: Fig-
ureSkating [19] and FineDiving [38]. FigureSkating com-
prises 11 videos featuring 371 short-program performances
from Winter Olympics and World Championships, anno-
tated with 4 event classes. Annotations cover 0.23% of the
frames. Evaluation is performed on two different splits: FS-
Comp and FS-Perf, as detailed in Hong et al. [20]. Fine-
Diving [38] consists of 3000 diving clips annotated with 4
different event classes, covering 2.2% of the frames. Both
datasets' videos have a frame rate between 25 and 30 frames
per second.

4.2. Evaluation

Following standard practices, we train on the train split, use
the validation split for early stopping, and evaluate on the
test split for all datasets. Model performance is assessed us-
ing the mAP metric, which calculates the mean of Average
Precisions across different events. We employ a tight ver-
sion of the metric with a tolerance of $? = 1$ frame and a
loose metric with $? = 2$ frames.

4.3. Implementation details

We train T-DEED using clips of $L = 100$ frames with a batch size of 8 clips. Each epoch comprises 5000 clips, randomly sampled from the training videos. Models are trained for 50 epochs using AdamW optimizer [26], with a base learning rate of $8e-04$, with 3 linear warmup epochs, and cosine decay. Positive classes in the cross-entropy loss are weighted with $w = 5$. We evaluate two versions of our feature extractor, denoted as RegNetY-200MF and RegNetY-800MF with hidden dimensions set to $d = 368$ and $d = 768$, respectively, and a downscaling factor of $k = 2$. Additional dataset-sensitive model hyperparameters are outlined in the supplementary material.

4.4. Comparison to SOTA

In Table 1, we compare our proposed T-DEED in two variations, with smaller and larger feature extractors, against previous SOTA models across four different configurations and datasets. We present mAP scores with $\tau = 1$ and $\tau = 2$. For the Figure Skating dataset, in both FS-Comp and FS-Perf configurations, our models exhibit improvements on the tight metric, with gains of up to +1.15 and +3.07, respectively. However, performance on the loose metric, with a 2-frame tolerance, is comparable for FS-Comp and slightly worse in FS-Perf. We attribute this to the use of dilation in the E2E-Spot model for the FigureSkating dataset, which may be beneficial when using larger tolerances. Additionally, in the FineDiving dataset, T-DEED clearly outperforms the SOTA on both metrics, with improvements of up to +4.87 on the tight metric and +3.58 on the loose metric.

4.5. Ablations

In this section, we incrementally ablate the different components of our approach. We start with a base model consisting of the feature extractor without GSF, producing features with only spatial information, and plain layers of the different temporal approaches. Initially, we consider only the classification head (i.e., $L = L_c$), resulting in per-frame classifications. For each approach, we evaluate multiple configurations with different hyperparameters (e.g., number of layers) and present results with the best configuration, reporting the mAP with a tolerance of $\tau = 1$. Each experiment is trained with two different seeds and we report the average for robustness. Ablations are conducted using two datasets: FineDiving and FigureSkating on the FS-Comp split.

Enhancing token discriminability. As previously discussed, tasks like PES require tokens with sufficient discriminability to prevent precision loss in predictions due to similar representations for spatially similar adjacent frames.

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Table 2. Ablation of T-DEED’s main components using mAP with $\tau = 1$, highlighting the best results in bold.

FS-Comp FineDiving

Experiments mAP ($\tau = 1$) mAP ($\tau = 1$)

(a) Temporal module

Transformer 69.68 59.84

GRU 72.04 59.09

SGP 73.14 60.07

(b) Skip connection

w/o 74.29 63.01

sum 73.27 60.14

concat 76.78 61.90

SGP-Mixer (sum) 76.88 61.34

Here, we evaluate the discriminability of three commonly used layers for modeling temporal information: Transformer [35], GRU [7], and SGP [31]. We measure the discriminability of their output tokens by averaging the cosine similarity between each token and the mean token of the sequence, as described in Shi et al. [31].

Figure 5 illustrates the rank-loss problem in Transformers, where token similarity increases across layers, therefore losing discriminability. In contrast, GRU layers, typically requiring fewer layers, show slight improvement in similarity. However, SGP layers exhibit the lowest similarity (i.e., highest discriminability) among output tokens, demonstrating its effectiveness in enhancing token discriminability. Additionally, this enhanced discriminability leads to the best performance among the different temporal modules, as shown in Table 2a. This observation highlights the advantages of employing the SGP layer, particularly in precision-demanding tasks like PES.

Defining multiple temporal scales. Employing multiple temporal scales while processing videos has proven effective across various TAL and AS methodologies [31, 32, 43]. Here, we evaluate our encoder-decoder architecture, operating on multiple temporal scales, using diverse mixture approaches in skip connections. Specifically, we assess five variations: (1) no skip connections, (2) addition, (3) concatenation and linear projection, (4) a modified SGP-Mixer aggregating branches information with summation, and (5) our proposed SGP-Mixer layer. Further details of each approach can be found in the supplementary material.

Results of these approaches are presented in Table 2b, where we observe that all encoder-decoder-based approaches, with or without skip connections, outperform the baseline using a single scale. This underscores the importance of capturing information from multiple temporal scales, effectively achieved in our encoder-decoder. However, further analysis reveals issues with approaches utilizing

(a) FS-Comp discriminability analysis.

(b) FineDiving discriminability analysis.

Figure 5. Temporal module discriminability analysis. Cosine similarity after backbone (BB), post-positional encoding (PE), and at each temporal layer is displayed. Additionally, mAP performance with $\tau = 1$ is reported.

lizing addition for feature aggregation within skip connections, as they tend to yield inferior results compared to alternative methods. We hypothesize that this problem may be due to the attribution of equal weight to features from skip connections and previous layers when adding them. This could be particularly critical in the top layers of the architecture, where information passed from skip connections may be too primitive. Finally, our proposed SGP-Mixer layer stands as the best approach for both datasets in aggregating information within the skip connections. This emphasizes the advantages of our carefully designed SGP-Mixer layer in aggregating information across multiple temporal scales while enhancing token discriminability.

Introducing the displacement head. In Table 3a we evaluate various methods for addressing the imbalance between frames containing actual events and background frames in PES. These methods include label dilation, which aims to detect events within a radius around the ground truth but may impact prediction precision, and the utilization of a displacement head, as detailed in Section 3.3. Results indicate that the prediction head offers more benefits compared to label dilation, enabling wider range of event detection with-

out sacrificing prediction precision.
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Table 3. Further ablations and analysis of T-DEED using mAP with $\gamma = 1$.

FS-Comp FineDiving

Experiments mAP ($\gamma = 1$) mAP ($\gamma = 1$)

(a) Displacement head

w/o & dilation = 0 74.74 63.67

w/o & dilation = 1 77.96 61.97

rE = 1 78.20 67.49

rE = 2 77.12 68.31

(b) Feature pyramids

Tridet 68.31 64.28

(c) Feature extractor

w/ gsm 81.00 67.95

w/ gsm (half) 81.05 67.47

w/ gsf 80.43 67.87

w/ gsf (half) 81.25 68.40

(d) Clip length

L = 25frames 71.02 66.61

L = 50frames 76.87 64.84

L = 100frames 81.25 68.40

L = 200frames 79.19 65.29

(e) Postprocessing

NMS [27] 81.27 68.40

SNMS [2] 85.15 71.48

Feature pyramids. A common approach in TAL to process multiple temporal scales is the use of feature pyramids, which resembles using only the encoder of our approach.

In Table 3b, we explore adapting the SGP Feature Pyramid proposed in Shi et al. [31] to our task. This method generates predictions at each temporal scale and integrates a displacement head to locate them. However, we observe a performance decrease compared to our encoder-decoder architecture. Further analysis of the predictions produced at each layer of the pyramid, depicted in Figure 6, reveals a decrease in performance as we descend the pyramid towards more high-level but lower-resolution features. This suggests that generating predictions at lower resolutions might compromise the accuracy. Our encoder-decoder architecture is able to model high-level information while recovering the original temporal resolution, thus avoiding losing the temporal precision that is critical in PES.

Feature extractor ablations. In Table 3c, we explore the impact of integrating different temporal shift modules into our 2D backbone to capture local temporal context. Specifically, we consider two modules, GSM [33] and GSF [34], known for their effectiveness in action recognition tasks. We assess two variants: one applying these modules across all backbone layers, and another limited to the latter half to promote stronger spatial modeling before temporal integration. Figure 6. Per-layer mAP analysis using the Feature Pyramid Network. mAP is reported at each layer of the SGP feature pyramid, accumulating predictions from previous layers, for the FineDiving and FigureSkating datasets.

gration. Following [20], we apply these modules to 1

4 of

the channels of the residual blocks. Results indicate that incorporating local temporal modeling into the backbone is especially useful in the FigureSkating dataset, while FineDiving shows comparable results to those without temporal modules. Notably, GSF applied to the latter half of the backbone yields the best performance for both datasets.

Clip length analysis. Regarding optimal clip length, Table 3d evaluates various number of frames. In FigureSkating, performance improves with increasing clip length, plateauing at 100 frames. Similarly, results for FineDiving also indicate 100 frames as the optimal choice.

Postprocessing analysis. In Table 3e, we illustrate the impact of employing two postprocessing techniques: Non-Maximum Suppression (NMS) [27] and Soft Non-Maximum Suppression (SNMS) [2]. The optimal results are achieved with a 1-frame window for NMS and a 3-frame window for SNMS. Notably, SNMS consistently improves mAP for both datasets.

5. Conclusion

This work presented T-DEED, a model designed to address Precise Event Spotting across various sports datasets. Ablation studies underscore the importance of processing videos in multiple temporal scales and enhancing token discriminability for precise predictions. To address these challenges, we integrate an encoder-decoder architecture and propose the SGP-Mixer layer, aimed at aggregating information at various temporal scales within skip connections while improving token discriminability. Additionally, T-DEED achieves SOTA performance on the FigureSkating and FineDiving datasets.

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12. assistant (2026-06-06T17:32:00.186Z)

I now have full content. This is a primary peer-reviewed research paper (CVPR 2024 Workshop, Computer Vision Foundation open access). Let me extract the falsifiable claims relevant to the research question.

Key relevance assessment: This paper is about Precise Event Spotting (PES) ? a temporal action localization method family. It's directly relevant to the TAL deep-model family in the research question, though notably it does NOT use badminton/racket-sport datasets (it uses FigureSkating and FineDiving). The license info (MIT-style GitHub code, but datasets are figure skating/diving) is relevant to commercial viability.

13. user (2026-06-06T17:32:11.821Z)

Structured output provided successfully

14. assistant (2026-06-06T17:32:21.248Z)

Extracted and returned the structured output. Summary of findings:

This is a **primary** peer-reviewed source (CVPR 2024 Workshop on Computer Vision in Sports, CVF open access; authors from Universitat de Barcelona, Computer Vision Center, and Aalborg University). I bypassed the failed WebFetch PDF extraction by reading the locally-saved PDF directly via pypdf.

Relevance to the research question: T-DEED is a concrete instance of the **Temporal Action Localization / Precise Event Spotting (PES)** deep-model family. Most relevant takeaways for the local-first, single-GPU pipeline:

- **Compute-friendly**: uses a compact 2D RegNetY backbone (200MF/800MF) with GSF temporal-shift modules, not heavy 3D nets ? plausible on an RTX 2070 Super.
- **Sparse labels suffice**: PES needs only per-keyframe annotations (0.23%?2.2% of frames labeled), which is a lighter labeling burden than dense interval segmentation ? notable given the n=2 data-starved situation.
- **No racket-sport data**: evaluated only on FigureSkating and FineDiving, so it offers no

badminton benchmark and would require relabeling/fine-tuning.

- ****License caveat****: code is on GitHub (github.com/arturxe2/T-DEED) but the paper states no explicit software license, and its datasets are figure-skating/diving ? not commercially-clear racket-sport data.

PDF saved at: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767069364-9guyic.pdf`